

114-3_2D Newton's Laws & Forces_



Useful Formulas for kinematics

$$(1) v = v_0 + at$$

$$(2) \Delta x = v_0 t + \frac{1}{2}at^2$$

$$(3) v^2 = v_0^2 + 2a\Delta x$$

Q1

A 20 kg table sits on a flat floor. The static friction coefficient is $\mu_s = 0.6$ and the kinetic friction coefficient is $\mu_k = 0.4$.

- (a) Calculate the normal force acting on the table.
- (b) You push the table with 100 N of force. Does the table move? What is the friction force pushing back against you right now?

Q2

Based on Q1, now you push harder with 150 N of force.

- (a) Which friction (static or kinetic) acts on it now?
- (b) Focus on the X-direction. Calculate the net force.
- (c) Calculate the acceleration of the table.

Q3

Two dogs are pulling a toy bone. Dog A pulls West with 60 N. Dog B pulls South with 80 N.

- (a) Draw a dot representing the toy bone, and draw these two force vectors acting on it.
- (b) Use the Pythagorean theorem to calculate the magnitude of the net force on the toy.

Q4

You are pulling a 50 kg sled on flat snow. Your pulling force is 100 N at an angle of 37° above the horizontal. The kinetic friction coefficient is $\mu_k = 0.1$.

- (a) Calculate the horizontal pull (F_x) and the vertical pull (F_y).
- (b) Calculate the normal force from the snow.
- (c) Find the net force in the X-direction.

Q5

A 50 kg skier is sliding down a ski slope tilted at 53° . The kinetic friction coefficient between the skis and the snow is $\mu_k = 0.5$.

- (a) Calculate the force parallel to the slope and the force perpendicular to the slope respectively.
- (b) What is the value of the normal force? Calculate the kinetic friction force.
- (c) Calculate the acceleration of the skier sliding down the slope.

Q6

Imagine a 20kg lawnmower.

- Situation A:

You **pull** with a force of 100 N at an angle 37° above the horizontal.

- Situation B:

You **push** with a force of 100 N at an angle 37° below the horizontal.

- (a) Draw the free-body-diagram (FBD) for both A and B situation. In which situation will the normal force be larger? Explain your reasoning based on the FBD.
- (b) Which situation (Pushing or Pulling) will have a larger friction force pushing back against you? Explain the reason.
- (c) Calculate the exact normal force for Situation B.